

Dataset Links

All sources are free. Every link below is drawn from the project brief and the master dataset reference. Click any blue label to open; the raw URL is shown beneath it for copy-paste.

CORE SOURCES — THE THREE PILLARS

1 · NVD CVE API 2.0 (NIST) — Pillar C, security

Primary security dataset — CVEs per vendor via CPE / `virtualMatchString`; CVSS severity + publication dates → per-vendor burden & severity trends.

[API endpoint \(JSON\)](https://services.nvd.nist.gov/rest/json/cves/2.0)

<https://services.nvd.nist.gov/rest/json/cves/2.0>

[Request a free API key](https://nvd.nist.gov/developers/request-an-api-key)

<https://nvd.nist.gov/developers/request-an-api-key>

[API documentation \(vulnerabilities\)](https://nvd.nist.gov/developers/vulnerabilities)

<https://nvd.nist.gov/developers/vulnerabilities>

⚠ Free key lifts the rate limit from 5 → 50 requests / 30s. Paginate, sleep ~1s, cache. CPE-based attribution is imperfect → validate on a sample and report attribution error.

2 · CISA KEV catalog — actively-exploited flags

Flags CVEs known to be actively exploited → per-vendor “exploited rate,” publication-to-KEV lag, ransomware-linked share.

[Known Exploited Vulnerabilities — JSON feed](https://www.cisa.gov/sites/default/files/feeds/known_exploited_vulnerabilities.json)

https://www.cisa.gov/sites/default/files/feeds/known_exploited_vulnerabilities.json

[Known Exploited Vulnerabilities — CSV feed](https://www.cisa.gov/sites/default/files/feeds/known_exploited_vulnerabilities.csv)

https://www.cisa.gov/sites/default/files/feeds/known_exploited_vulnerabilities.csv

[Catalog homepage \(human-readable\)](https://www.cisa.gov/known-exploited-vulnerabilities-catalog)

<https://www.cisa.gov/known-exploited-vulnerabilities-catalog>

› No key, no rate limit; the GitHub mirror is CC0. Normalize the vendorProject / product name fields.

3 · SEC EDGAR XBRL — Pillar A, financials

Primary financial dataset — total & reportable-segment revenue / operating income per vendor; the revenue-mix view; demonstrates the AWS-vs-Azure/GCP disclosure asymmetry.

[Company Facts API \(example: Microsoft, CIK 0000789019\)](https://data.sec.gov/api/xbrl/companyfacts/CIK#####.json)

<https://data.sec.gov/api/xbrl/companyfacts/CIK#####.json>

[CIK ↔ ticker lookup map](https://www.sec.gov/files/company_tickers.json)

https://www.sec.gov/files/company_tickers.json

⚠ Requires a User-Agent: name email header; 10 requests/sec. us-gaap tags don't always expose the cloud sub-segment → you may need to parse the 10-K segment note.

4 · Stack Overflow Annual Developer Survey — Pillar B

Adoption corroboration — YoY share of respondents using each cloud / DB / AI tool vs. the revenue-momentum ranking.

[Survey results & raw data downloads](https://survey.stackoverflow.co/)

<https://survey.stackoverflow.co/>

⚠ Free (respect ODbL). Columns change yearly → harmonize; the sample is self-selected → treat as directional only.

5 • GH Archive

Pillar B enrichment — star / push activity for AI-framework & cloud repos as a mindshare proxy.

[GH Archive](https://www.gharchive.org/)

<https://www.gharchive.org/>

⚠ Query via BigQuery within the free 1 TB/month tier — never bulk-download the TB-scale history.

6 • MITRE cvelistV5 — offline CVE fallback

Bulk CVE validation / gap-fill when the NVD API is rate-limited.

[CVEProject/cvelistV5 \(GitHub\)](https://github.com/CVEProject/cvelistV5)

<https://github.com/CVEProject/cvelistV5>

› Large repo — shallow / sparse clone. Keep secondary to NVD to avoid double-counting.

7 • yfinance (secondary)

Equity price panel for the exploratory KEV-spike-vs-stock analysis + a valuation sanity check.

[yfinance library \(GitHub\)](https://github.com/ranaroussi/yfinance)

<https://github.com/ranaroussi/yfinance>

› Fragile — cache to Parquet. This analysis is explicitly non-causal — label it as such.